Torque-Aware Tuning of Fractional-Order PID Controllers for a Six-Degree-of-Freedom Robotic Arm with a Fractional-Order PSO–Grey Wolf Hybrid

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Abstract. Fractional-order PID (FOPID) control of a robotic arm requires five parameters per joint, and tuning for tracking error alone can produce controllers that demand excessive torque. This paper tunes the six independent FOPID controllers (30 parameters) of a six-degree-of-freedom arm with FOPSO-GWO, a hybrid optimiser that combines the fractional-order velocity memory of fractional PSO with a grey-wolf pull towards the three best solutions. The objective scores step and sinusoidal tracking together with the control effort, subject to 24 per-joint limits on peak torque and overshoot. All eight tuning runs satisfy every limit. Against PID and FOPID baselines matched to published results for the same arm, the tuned controller lowers the mean overshoot by 73 % and 53 %, the sinusoidal mean squared error by 91 % and 77 % and the peak torque of the derivative kick by about 79 %. Tuning for the ITAE alone breaks 7 of the 18 torque limits.

Keywords: fractional-order PID, robotic manipulator, particle swarm optimisation, grey wolf optimiser, trajectory tracking, control effort

1 Introduction

Robotic manipulators are multi-input, multi-output systems with strongly coupled, nonlinear dynamics. In industry their joints are still mostly controlled by independent PID loops, which are simple, model-free and well understood [1]. The fractional-order PID (FOPID) controller, $\mathrm{PI}^{\lambda}\mathrm{D}^{\mu}$, generalises the PID by letting the orders of integration λ and differentiation μ take non-integer values [13]. The two extra degrees of freedom shape the frequency response more flexibly than an integer controller can, at the price of a harder tuning problem [10, 14, 18]. A six-joint arm has $6 \times 5 = 30$ FOPID parameters, and the joints interact through the inertia and Coriolis couplings of the arm. The tuning problem is therefore high-dimensional, non-convex and can only be evaluated by simulation, which is why population-based metaheuristics are commonly used for it.

Swarm and nature-inspired algorithms have tuned FOPID controllers for wind energy conversion with the grey wolf optimiser (GWO) [20] and for load frequency control with the whale optimisation algorithm [7]. GWO has also been used to plan the motion of 6-DOF manipulators [2]. Particle swarm optimisation (PSO) [6, 15] has been extended with a fractional-order velocity, which gives every particle a memory of its recent velocities [12, 3]. It has also been hybridised with GWO, whose leader-based update draws the swarm towards several good solutions at once rather than towards a single global best [9, 17].

Most of these studies minimise one tracking integral, typically the integral of the time-weighted absolute error (ITAE). For a multi-joint arm this is not enough. A tracking-only objective rewards stiff, high-gain controllers, whatever torque they demand. On a step reference, the derivative action also produces a short but very large torque *kick*. A controller that is optimal for the ITAE may therefore be unusable on real actuators. Section 6.4 shows this for the arm studied here: an ITAE-tuned FOPID exceeds 7 of 18 per-joint torque limits, with a derivative kick 14 times that of the controller proposed here.

This paper makes three contributions:

1. A *whole-controller* tuning objective. It scores step and sinusoidal tracking together with four measures of control effort, subject to 24 per-joint limits on peak torque and overshoot, and includes a check that rejects numerically ill-conditioned loops.
2. FOPSO-GWO, a fractional-order PSO with a grey-wolf term. Its settings were selected on development seeds that are kept separate from the reported runs, and an ablation shows what each component contributes.
3. An evaluation on a UR5-type 6-DOF arm, with PID and FOPID baselines that reproduce published results for the same arm [5] within 6.5 %. The evaluation covers joint-by-joint tracking, control effort and the spread over eight independent runs.

Section 2 describes the arm, the controller and the test scenarios; Sect. 3, the baseline controllers; Sect. 4, the tuning problem; and Sect. 5, FOPSO-GWO. Section 6 presents the results, Sect. 7 discusses limitations, and Sect. 8 concludes.

2 Arm Model, Controller and Test Scenarios

2.1 Manipulator Dynamics

The arm is a UR5-type manipulator with six revolute joints. It follows the standard Denavit–Hartenberg (DH) convention, with the link lengths, masses and principal inertias of the benchmark in [5] (Table 1). Its joint-space dynamics are [16]

$$M(q)\,\ddot{q} + C(q,\dot{q})\,\dot{q} + G(q) + \tau_f(\dot{q}) = \tau, \tag{1}$$

where $q \in \mathbb{R}^6$ are the joint angles, M is the inertia matrix, $C\dot{q}$ collects the Coriolis and centrifugal torques, G the gravity torques, τ_f the friction torques

Table 1. Arm parameters: standard DH parameters, link masses and principal inertias about the centre of mass [5]

Joint i	d_i (m)	a_i (m)	α_i (rad)	m_i (kg)	$(I_{xx}, I_{yy}, I_{zz})_i$ (kg m^2)
1	0.128	0	$\pi/2$	2.0	(0.001, 0.001, 0.001)
2	0	−0.612	0	2.5	(0.040, 0.040, 0.040)
3	0	−0.571	0	5.7	(0.060, 0.060, 0.040)
4	0.164	0	$\pi/2$	3.9	(0.055, 0.055, 0.040)
5	0.115	0	$-\pi/2$	2.5	(0.040, 0.040, 0.040)
6	0.092	0	0	2.5	(0.040, 0.040, 0.040)

and τ the joint torques. M, $C\dot{q}$ and G are computed by the recursive Newton–Euler algorithm [8]. The implementation was checked in two ways. Its inertia matrix agrees with an independent Jacobian-based formula to 10^{-15}, and the simulated unforced motion conserves energy to 10^{-14}. The centres of mass are not reported in [5], so they are placed at the origins of the DH frames. The published responses of the benchmark stay at rest until the reference changes. Gravity and joint friction are therefore omitted ($G = 0$, $\tau_f = 0$), consistent with that scenario. The model is integrated with a fourth-order Runge–Kutta scheme with a 1 ms step.

2.2 Fractional-Order PID Control

Each joint j has its own FOPID controller acting on the tracking error $e_j = q_{r,j} - q_j$:

$$\tau_j(t) = K_{p,j}\, e_j(t) + K_{i,j}\, D^{-\lambda_j} e_j(t) + K_{d,j}\, D^{\mu_j} e_j(t), \qquad j = 1, \ldots, 6, \tag{2}$$

where D^{α} is the fractional operator of order α. The PID is the special case $\lambda_j = \mu_j = 1$. The controllers run at 1 kHz with a zero-order hold. Orders above one are split into an integer part, realised as an integrator or a backward difference, and a fractional remainder $\alpha \in (0, 1)$. The remainder is realised by Oustaloup's recursive approximation [11],

$$s^{\alpha} \approx \omega_h^{\alpha} \prod_{k=-N}^{N} \frac{s + \omega_k'}{s + \omega_k}, \quad \omega_k' = \omega_b \Big(\tfrac{\omega_h}{\omega_b}\Big)^{\frac{k+N+(1-\alpha)/2}{2N+1}}, \quad \omega_k = \omega_b \Big(\tfrac{\omega_h}{\omega_b}\Big)^{\frac{k+N+(1+\alpha)/2}{2N+1}}, \tag{3}$$

with $N = 5$ over the band $[\omega_b, \omega_h] = [10^{-3}, 10^{3}]$ rad/s. The filter is expanded into partial fractions so that each first-order section can be discretised exactly with a zero-order hold. This keeps the fastest poles, near 10^3 rad/s, stable at the 1 ms sampling period.

2.3 Test Scenarios and Performance Metrics

The controllers are evaluated in the two scenarios of [5], each lasting 5 s with the arm starting at rest at $q = 0$:

Table 2. Baseline controllers: values reported in [5] and the reproduction used here, with the relative deviation

		σ (%)	t_s (s)	t_p (s)	MSE (rad^2)	$\Sigma\lvert\tau\rvert$ (N m)
PID	reported [5]	54.6	2.44	1.39	2.27×10^{-2}	3.74×10^{4}
	reproduced	51.1	2.49	1.45	2.12×10^{-2}	3.50×10^{4}
	deviation	−6.5 %	+2.2 %	+4.6 %	−6.5 %	−6.5 %
FOPID	reported [5]	31.2	1.89	1.33	8.80×10^{-3}	2.57×10^{4}
	reproduced	29.2	1.95	1.35	8.29×10^{-3}	2.41×10^{4}
	deviation	−6.4 %	+3.1 %	+1.6 %	−5.8 %	−6.1 %

- a *step* of 1 rad applied to every joint at $t = 1$ s;
- *sinusoidal tracking* of $q_{r,j}(t) = \sin(1.5\,t)$ rad on every joint.

Signals are logged every 0.01 s. Five metrics are averaged over the six joints:

- the step overshoot σ;
- the adjustment (settling) time t_s, the last exit from a ± 5 % band;
- the peak time t_p;
- the mean squared error MSE of the sinusoidal run;
- the sum of absolute torques $\Sigma|\tau|$ over the 501 logged samples of the sinusoidal run.

As in [5], t_s and t_p are measured from $t = 0$, so both are at least 1 s. Tracking over time is summarised by

$$\mathrm{ITAE} = \int_0^5 t \sum_{j=1}^{6} |e_j(t)|\, dt, \tag{4}$$

evaluated for each scenario.

3 Baseline Controllers

The benchmark [5] reports the performance of a PID and a FOPID controller in both scenarios, but not their gains. For a like-for-like comparison, gains were identified for both controllers, one set per scenario. Every metric of the reproduced controllers lies within 6.5 % of the reported value (Table 2).

The identification used CMA-ES [4]. The deviation from each reported metric was a constraint, enforced as a penalty above a tolerance. Within that tolerance, the search minimised the root-mean-square difference between the simulated joint trajectories and the published response curves. Candidates were rejected if their response changed by more than 10^{-6} rad when every gain was perturbed by a relative 10^{-10}, which marks a numerically ill-conditioned loop. Candidates whose torque chattered were also rejected.

4 The Tuning Problem

4.1 Decision Variables

The optimiser tunes all 30 parameters at once, because the joints are dynamically coupled. Each parameter is mapped to $[0, 1]$:

- the gains on a logarithmic scale, with $\log_{10} K_p \in [-2, 5]$, $\log_{10} K_i \in [-4, 5]$ and $\log_{10} K_d \in [-1, 3]$;
- the orders linearly, with $\lambda, \mu \in [0.05, 1.95]$.

One gain set is used for both scenarios.

4.2 Whole-Controller Objective

A candidate controller is simulated in both scenarios, at the 1 kHz control rate for the torque terms. Its metrics are compared with reference values $\bar{m}_i$ through the ratios

$$\rho_1 = \frac{\mathrm{ITAE}_{\mathrm{step}}}{\overline{\mathrm{ITAE}}_{\mathrm{step}}}, \quad \rho_2 = \frac{\mathrm{ITAE}_{\mathrm{sine}}}{\overline{\mathrm{ITAE}}_{\mathrm{sine}}}, \quad \rho_3 = \frac{\sigma}{\bar{\sigma}}, \quad \rho_4 = \frac{t_s - 1}{\bar{t}_s - 1},$$
$$\rho_5 = \frac{t_p - 1}{\bar{t}_p - 1}, \quad \rho_6 = \frac{\mathrm{MSE}}{\overline{\mathrm{MSE}}}, \quad \rho_7 = \frac{\Sigma|\tau|}{\overline{\Sigma|\tau|}}. \tag{5}$$

Inside the objective, the times are measured from the step instant. Otherwise every time would be at least 1 s and ρ_4, ρ_5 would hardly change.

Four effort ratios $\epsilon_1, \ldots, \epsilon_4$ compare $\Sigma|\tau|$ and the total variation of the torque, $\mathrm{TV} = \sum_k |\tau(k+1) - \tau(k)|$, in both scenarios with their reference values. The total variation grows with chattering and with noise amplification. In the step scenario, the first 50 ms after the step (the derivative kick) are excluded from the effort ratios and limited separately.

The objective is

$$J_W = \frac{\sum_{i=1}^{6} w_i \max(\rho_i, 0.5) + \sum_{k=1}^{4} \epsilon_k}{\sum_{i=1}^{6} w_i + 4} + 2 \sum_{i=3}^{7} \max(0, \rho_i - 1) + 10\, P_{\mathrm{cap}} + 10\, [\text{ill-conditioned}], \tag{6}$$

with weights $w = (0.5, 0.5, 1, 1, 1, 1)$. The terms work as follows:

- The floor at 0.5 gives no credit for being more than twice as good as the reference on any one metric. Without it, a controller can trade a large loss on one metric for an ever-larger gain on another.
- The second term penalises every reported metric (ρ_3 to ρ_7) that does not improve on its reference value.

Table 3. Constants of the objective J_W: reference values (top) and per-joint limits (bottom)

$\bar{\sigma}$	22.1 %	$\overline{\mathrm{ITAE}}_{\mathrm{step}}$, $\overline{\mathrm{ITAE}}_{\mathrm{sine}}$	1.158, 2.440
$\bar{t}_s$, $\bar{t}_p$	1.43 s, 1.09 s	step $\Sigma\lvert\tau\rvert$, sine $\Sigma\lvert\tau\rvert$ (N m)	6343, 10145
$\overline{\mathrm{MSE}}$	3.70×10^{-3} rad^2	step TV, sine TV (N m)	934.7, 1444.8
$\overline{\Sigma\lvert\tau\rvert}$	2.32×10^4 N m		

Peak torque (N m)	J1	J2	J3	J4	J5	J6
kick window	1.11×10^6	2.11×10^5	1.50×10^5	1.09×10^4	4.55×10^4	5.08×10^4
step after kick	1524	1495	754.3	159.7	60.9	639.0
sinusoidal run	1669	1902	627.2	495.1	480.8	64.3
overshoot (%)	33.1	33.1	33.1	33.1	33.1	33.1

- P_{cap} sums the relative excess, $\max(0, x/x_{\max} - 1)$, over 24 per-joint limits. On each joint, the peak torque in the kick window, in the rest of the step run and in the sinusoidal run may not exceed the largest peak that any of the reference controllers identified from the published responses of [5] needs on that joint. No joint may overshoot by more than 33.1 %, a limit also taken from those published responses.
- The last term applies the conditioning check of Sect. 3.

Simulations are aborted as soon as any joint error exceeds 5 rad. Such candidates receive a cost above 10^3 that decreases with the time they survived.

The reference values $\bar{\sigma}, \bar{t}_s, \bar{t}_p, \overline{\mathrm{MSE}}, \overline{\Sigma|\tau|}$ are those reported for this arm in [5]. The two ITAE references and the four effort references were computed once, from a FOPID identified from the published responses. Table 3 lists all constants.

4.3 Tracking-Only Objectives for Comparison

Two tracking-only objectives are used to show what the effort terms and limits change, and for the ablation in Sect. 6.5. Both keep the divergence handling and the conditioning check.

- J_T is the weighted mean of $\rho_1, \ldots, \rho_7$, with weights $(0.5, 0.5, 1, 1, 1, 1, 1)$, plus the same penalty on reported metrics that do not improve. The two ITAEs are taken relative to the start controller, and the times are measured from the step as in J_W. There are no floor, effort terms or limits.
- J_{ITAE} is the step ITAE relative to that of the start controller, the classical single-criterion objective.

5 FOPSO-GWO

FOPSO-GWO keeps the structure of PSO [6, 15] and changes the velocity update in two ways. First, the velocity carries a fractional-order memory of the last four

Algorithm 1 FOPSO-GWO

Require: objective J, start controller $z_0 \in [0,1]^{30}$, swarm size N, iterations K

$x_1 \leftarrow z_0$; $x_2, \ldots, x_{0.3N+1} \leftarrow \mathrm{clip}(z_0 + 0.05\,\mathcal{N}(0, I))$; remaining particles uniform on $[0,1]^{30}$
evaluate $J(x_i)$; $p_i \leftarrow x_i$; $g \leftarrow \arg\min_i J(p_i)$; all velocities $\leftarrow 0$
for $k = 1, \ldots, K$ **do**
    $w_k \leftarrow 0.9 - 0.5\,k/K$; $a_g \leftarrow 2(1 - k/K)^2$
    $\alpha, \beta, \delta \leftarrow$ the three best personal bests
    for each particle i **do**
        compute $\hat{x}_i$ by (8) and v_i^{k+1} by (7); clamp $|v_i^{k+1}| \le 0.2$
        $x_i \leftarrow \mathrm{clip}(x_i + v_i^{k+1}, 0, 1)$; zero the velocity components that hit a bound
        evaluate $J(x_i)$; update p_i if improved
    end for
    $g \leftarrow \arg\min_i J(p_i)$
end for
return g

velocities, obtained from the Grünwald–Letnikov expansion of a derivative of order a [12]. Second, a third attraction pulls each particle towards a grey-wolf estimate built from the three best personal bests [9]. For particle i at iteration k,

$$\begin{aligned} v_i^{k+1} = {} & (w_k - 1 + a)\, v_i^k + \tfrac{a(1-a)}{2}\, v_i^{k-1} + \tfrac{a(1-a)(2-a)}{6}\, v_i^{k-2} \\ & + \tfrac{a(1-a)(2-a)(3-a)}{24}\, v_i^{k-3} \\ & + c_1 r_1 \odot (p_i - x_i^k) + c_2 r_2 \odot (g - x_i^k) + c_3 r_3 \odot (\hat{x}_i - x_i^k), \end{aligned} \tag{7}$$

$$\hat{x}_i = \frac{1}{3} \sum_{L \in \{\alpha, \beta, \delta\}} \big(L - A_L \odot |C_L \odot L - x_i^k|\big), \tag{8}$$

with $A_L = 2a_g r_L - a_g$ and $C_L = 2r'_L$. The symbols are:

- p_i is the personal best of particle i and g the global best.
- α, β, δ are the three best personal bests.
- r_1, r_2, r_3, r_L, r'_L are vectors of independent uniform random numbers on $[0,1]$, and $\odot$ is the element-wise product.
- The inertia w_k falls linearly from 0.9 to 0.4 [15].
- The grey-wolf coefficient $a_g = 2(1 - k/K)^2$ falls from 2 to 0 over the K iterations. While $|A_L| > 1$, the wolf term throws particles past the leaders (exploration). As $a_g \to 0$, it contracts onto the three leaders (exploitation).

Velocities are clamped to ± 0.2 of each range. Positions are clipped to the unit box, and the velocity of a component that hits a bound is set to zero. Algorithm 1 summarises the method.

Table 4 lists the settings. Two choices differ from the usual practice of adding the wolf term on top of standard PSO:

Table 4. Settings of FOPSO-GWO

Swarm size, iterations, evaluations	$N = 30$, $K = 100$, $N(K + 1) = 3030$
Initial swarm	start, 9 jittered copies ($\sigma = 0.05$), 20 uniform
Attraction coefficients	$c_1 = c_2 = 1.5$, $c_3 = 1$
Fractional order	$a = 0.9$ (constant)
Inertia weight	w_k: $0.9 \to 0.4$, linear
Grey-wolf coefficient	$a_g = 2(1 - k/K)^2$
Velocity clamp	± 0.2 of each range

- The coefficients $c_1 = c_2 = 1.5$ and $c_3 = 1$ split the total attraction $c_1 + c_2 = 4$ of standard PSO between the PSO and grey-wolf terms. Adding c_3 on top of $c_1 = c_2 = 2$ made the swarm erratic.
- The fractional order is held at $a = 0.9$. With the order falling from 0.9 to 0.4, the total weight on the remembered velocities in (7) falls from 0.86 to 0.03, and the swarm loses its momentum long before the end of the run.

These settings were selected among 13 candidates on four development random seeds (101–104). Those seeds are never used for the results reported here, and the selection rule was fixed before the last candidates were run. The swarm always contains the start controller, so the result is never worse than it.

6 Results

6.1 Protocol

FOPSO-GWO was run eight times for each objective, with random seeds 1–8 and 3030 evaluations per run. Each evaluation of J_W simulates both scenarios, and repeats both for the conditioning check. All runs start from the same FOPID, identified from the published FOPID responses of [5] with one gain set for both scenarios. This start controller has $\sigma = 35.9\,\%$, $t_s = 2.10\,\mathrm{s}$, $t_p = 1.32\,\mathrm{s}$, MSE $= 1.05 \times 10^{-2}\,\mathrm{rad}^2$ and $J_W = 35.4$. Most of that cost comes from exceeded limits. The reported controller, FOPSO-GWO-FOPID, is the best of the eight runs under J_W. The simulation loop was compiled to C, so each evaluation takes a fraction of a second and a run takes about six minutes.

6.2 Convergence and Run-to-Run Spread

Figure 1(a) shows the best J_W of each run against the number of evaluations, and Table 5 gives the final values:

- Every run reduces J_W from 35.4 to between 1.42 and 2.16.
- All eight final controllers satisfy all 24 limits.
- Each final controller ends at or just below at least one torque limit (98–100 % of it), so the limits are active constraints.

Table 5. Final objective values of FOPSO-GWO over eight independent runs (seeds 1–8); lower is better

Objective	start	best	median	mean	worst
J_W (whole controller)	35.4	1.416	1.718	1.764	2.160
J_T (tracking only)	–	0.331	0.348	0.352	0.384
J_{ITAE} (step ITAE)	1	0.0865	0.1218	0.1160	0.1346

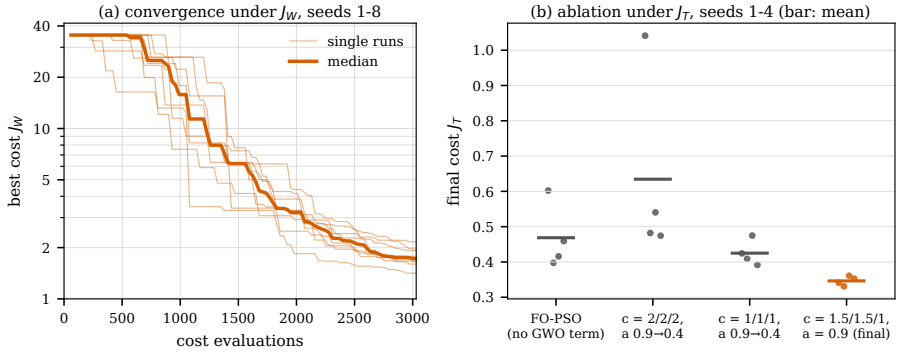


Fig. 1. (a) Best J_W against cost evaluations for the eight runs, and their median. (b) Ablation: final J_T of four variants on seeds 1–4 (Sect. 6.5).

– The median run reaches $J_W = 5$ after about 1650 evaluations and $J_W = 2$ after about 2600. All eight runs still improved during their last ten iterations, so a larger budget would probably improve the results further.

6.3 Tracking Performance

Table 6 compares FOPSO-GWO-FOPID with the two baselines, and Figs. 2 and 3 show the responses joint by joint.

– **Step.** The tuned controller lowers the mean overshoot from 51.1 % (PID) and 29.2 % (FOPID) to 13.6 %. The adjustment time falls to 1.35 s and the peak time to 1.11 s. Measured from the step instant, these times fall by 76 % and 75 % against the PID, and by 63 % and 68 % against the FOPID.
– **Sinusoid.** The MSE falls by 91 % and 77 %.
– **ITAE.** Both ITAEs fall by 45–73 %.

The averages hide differences between the joints (Table 7).

– **Largest gains: joints 2, 5 and 6.** The overshoot of joint 5 falls from 150.2 % (PID) and 93.5 % (FOPID) to 6.7 %. Its sinusoidal MSE falls by more than two orders of magnitude.

Table 6. Joint-averaged performance of the baselines and the tuned controller, and the relative change achieved by FOPSO-GWO-FOPID (negative is better)

Metric	PID	FOPID	FOPSO-GWO-FOPID	vs PID	vs FOPID
Overshoot σ (%)	51.1	29.2	13.6	−73.3 %	−53.4 %
Adjustment time t_s (s)	2.49	1.95	1.35	−45.8 %	−30.6 %
Peak time t_p (s)	1.45	1.35	1.11	−23.5 %	−17.8 %
Sine MSE (rad^2)	2.12×10^{-2}	8.29×10^{-3}	1.88×10^{-3}	−91.2 %	−77.4 %
Sine $\Sigma\lvert\tau\rvert$ (N m)	3.50×10^{4}	2.41×10^{4}	9.61×10^{3}	−72.5 %	−60.2 %
ITAE, step	4.34	2.16	1.19	−72.6 %	−45.1 %
ITAE, sine	5.81	4.74	1.72	−70.4 %	−63.7 %

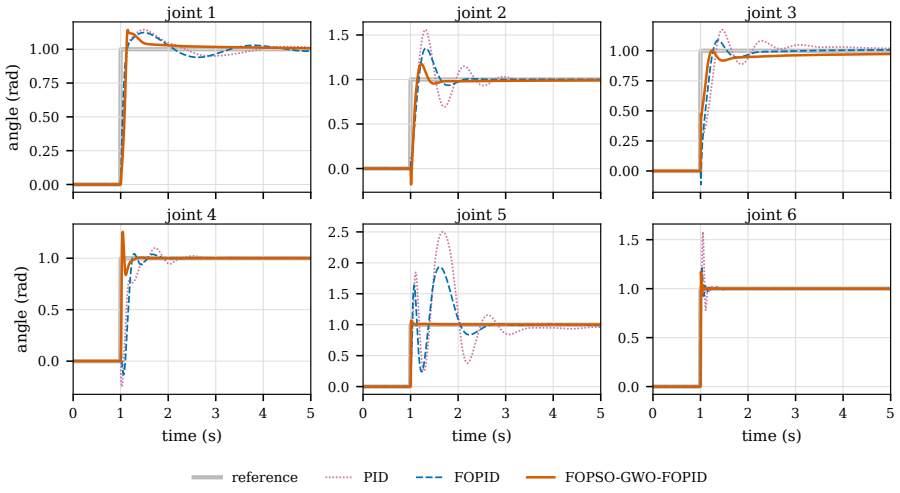


Fig. 2. Step responses of the six joints (1 rad at $t = 1$ s).

- **Joint 4 is worse.** Its overshoot rises to 25.7 %, against 10.0 % and 4.3 %. It still stays within the 33.1 % limit.
- **Joint 3 is overdamped.** It hardly overshoots (0.1 %), but settles in 2.07 s, slower than with the FOPID (1.87 s).
- **Joint 1 hardly changes.** Its overshoot (14.3 %) is close to that of both baselines, and it keeps the largest sinusoidal error of all joints (Fig. 3).

These trade-offs follow from the objective, which scores joint averages and limits each joint only through the caps.

Table 8 lists the tuned parameters. Some of them lie at or near the bounds of the search box: $\lambda_6 = 0.050$, and $K_{d,2} = 989$ against an upper bound of 1000. A wider box might therefore allow further improvement.

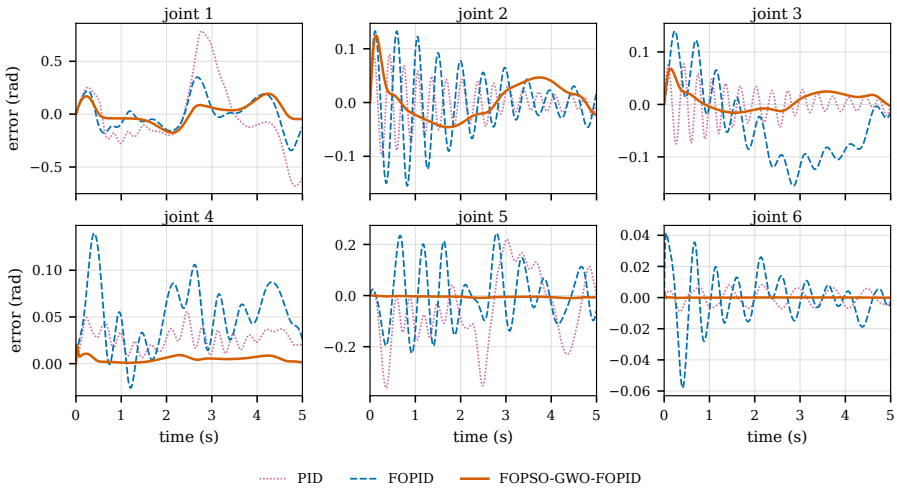


Fig. 3. Tracking error $e_j = q_{r,j} - q_j$ for the sinusoidal reference $q_{r,j} = \sin(1.5\,t)$ rad.

6.4 Control Effort and the Role of the Objective

Table 9 reports the torque at the 1 kHz control rate. Peaks are the largest over the joints; sums and total variations are averaged over the joints.

FOPSO-GWO-FOPID improves several effort measures:

- The derivative kick falls from about 2.2×10^5 N m to 4.6×10^4 N m, a 79 % reduction.
- The peak torque of the sinusoidal run falls by 53 % and 20 % against the PID and FOPID.
- The total variation of the sinusoidal torque falls by 91 % and 80 %.

Two step-run measures are not better:

- The peak torque after the kick is about that of the PID (1517 against 1503 N m).
- The total variation of the step torque is 45 % higher than for either baseline.

Perfectly tracking the sinusoid requires $\Sigma|\tau| = 7.40 \times 10^3$ N m by inverse dynamics; the tuned controller uses 1.30 times that. The 60–73 % reduction of $\Sigma|\tau|$ in Table 6 should be read with care. The baseline gains were fitted to the torque sums *reported* in [5]. PID and FOPID gains fitted to the published response curves instead need only about 1.0×10^4 N m.

The last column of Table 9 shows what the effort terms and the limits buy. It is the best of the eight FOPSO-GWO runs tuned for J_{ITAE} alone.

- **Tracking is better.** Its step ITAE is 0.248 against 1.188, and its sinusoidal MSE is 3.1×10^{-4} against 1.9×10^{-3} rad^2.

Table 7. Per-joint step overshoot, adjustment time and peak time, and sinusoidal MSE ("Tuned": FOPSO-GWO-FOPID)

Metric	Controller	J1	J2	J3	J4	J5	J6
σ (%)	PID	14.5	56.0	17.5	10.0	150.2	58.1
	FOPID	12.4	35.1	8.8	4.3	93.5	21.1
	Tuned	14.3	17.6	0.1	25.7	6.7	17.3
t_s (s)	PID	1.99	2.64	2.47	2.07	4.66	1.14
	FOPID	2.85	1.90	1.87	1.48	2.52	1.08
	Tuned	1.46	1.34	2.07	1.18	1.03	1.03
t_p (s)	PID	1.49	1.31	1.46	1.73	1.68	1.05
	FOPID	1.49	1.32	1.37	1.29	1.61	1.03
	Tuned	1.15	1.22	1.24	1.04	1.01	1.01
MSE (rad^2)	PID	1.0e−1	1.2e−3	8.4e−4	8.2e−4	2.1e−2	2.2e−5
	FOPID	2.3e−2	3.7e−3	6.7e−3	3.8e−3	1.2e−2	2.7e−4
	Tuned	9.4e−3	1.4e−3	4.0e−4	3.4e−5	3.3e−5	3.0e−8

Table 8. Parameters of FOPSO-GWO-FOPID (one set for both scenarios)

	J1	J2	J3	J4	J5	J6
K_p	110.7	0.2054	0.03346	590.0	33.96	3393
K_i	42.81	132.5	0.7265	43.96	4.420	4.329×10^{-4}
K_d	156.1	989.1	890.0	9.623	171.5	0.3650
λ	1.534	0.07579	1.377	0.07649	0.5251	0.05009
μ	0.8239	0.5172	0.5289	0.9728	0.7322	1.637

- **Effort is far worse.** Its derivative kick reaches $6.5 \times 10^5\,\mathrm{N\,m}$, 14 times that of FOPSO-GWO-FOPID. Its peak sinusoidal torque is 3.4 times higher.
- **It is not admissible.** It exceeds 7 of the 18 torque limits: the kick limit on two joints, up to 14.9 times, and the step limit on five joints, up to 1.9 times.

The whole-controller objective gives up some tracking to stay within limits that an actuator can meet.

6.5 Ablation

Table 10 and Fig. 1(b) compare FOPSO-GWO with variants that each change one design choice, all under the same budget and initial swarms.

- **Grey-wolf term.** Removing it, which leaves a fractional-order PSO with $c_1 = c_2 = 2$, raises the mean J_T from 0.346 to 0.469 (+35 %).
- **Coefficients.** Adding the wolf term on top of $c_1 = c_2 = 2$ is worse still (0.634), because the swarm becomes erratic. With $c = 1/1/1$ and a decreasing order, it collapses early (0.425).

Table 9. Control effort at the 1 kHz control rate (N m). The last column is the best run of FOPSO-GWO tuned for J_{ITAE} alone

	PID	FOPID	FOPSO-GWO-FOPID	ITAE-tuned
Kick peak (first 50 ms)	2.16×10^5	2.28×10^5	4.64×10^4	6.48×10^5
Step peak after the kick	1503	3114	1517	2797
Sine peak	879.2	516.7	412.9	1417
Step $\Sigma\lvert\tau\rvert$ after the kick	1.22×10^4	8928	8480	9012
Sine $\Sigma\lvert\tau\rvert$	3.50×10^4	2.41×10^4	9610	9512
Step total variation	1054	1072	1552	1665
Sine total variation	6307	2940	596.6	1339
Limits exceeded (of 18)	–	–	0	7

Table 10. Ablation: mean final cost of FOPSO-GWO variants under the same budget and initial swarms (J_T and J_{ITAE}: seeds 1–4; J_W: seeds 1–8)

Variant	$c_1/c_2/c_3$	order a	J_T	J_W	J_{ITAE}
Fractional-order PSO (no grey-wolf term)	2/2/0	$0.9 \to 0.4$	0.469	–	–
Wolf term added on top	2/2/2	$0.9 \to 0.4$	0.634	–	–
Initial settings	1/1/1	$0.9 \to 0.4$	0.425	1.929	0.116
FOPSO-GWO (final)	1.5/1.5/1	0.9	**0.346**	**1.764**	0.123

- **Consistency under J_W.** The final settings lower the median of J_W from 1.951 to 1.718 and its worst run from 2.694 to 2.160.
- **Where the initial settings win.** Their single best J_W run is lower (1.076 against 1.416). Under J_{ITAE} their mean is marginally lower (0.116 against 0.123 on seeds 1–4).

The final settings therefore trade some peak performance for consistency, which is the property that matters when only a few runs can be afforded.

7 Discussion and Limitations

The results support a narrower claim than "FOPSO-GWO is a better optimiser". Tuning the 30 FOPID parameters with the whole-controller objective yields a controller that clearly outperforms the reproduced PID and FOPID baselines, while respecting per-joint torque limits that an ITAE-tuned controller violates. Several limitations qualify this result:

- **Baselines.** They reproduce published results; they were not re-tuned with the same objective and budget. The comparison therefore measures the benefit of tuning with J_W, not the merit of FOPSO-GWO relative to other optimisers. Such a claim requires a comparison with other tuning algorithms under identical budgets, initial populations and objectives, and no algorithm can be expected to win on every problem [19].

- **Statistics.** Eight runs per objective are enough to show the spread of the method, but not to resolve small differences between variants. Under J_{ITAE} the seed-to-seed spread is as large as the differences between settings.
- **Model.** The arm is simulated with nominal parameters, without gravity, joint friction, actuator dynamics, sensor noise or torque saturation, and only for the two reference signals of the benchmark. Robustness to parameter errors and experiments on hardware remain to be shown.
- **Limits.** The tuned controller operates close to several limits. In practice the limits should be set with a margin below the actuator ratings.

The objective is close to a sum of per-joint contributions, because each joint's controller mostly shapes its own response. This suggests decomposing the search by joint as a direction for future work.

8 Conclusion

This paper tuned the six FOPID controllers of a 6-DOF robotic arm with FOPSO-GWO, a fractional-order particle swarm optimiser with a grey-wolf term. The whole-controller objective balances step and sinusoidal tracking against control effort under 24 per-joint limits. Compared with PID and FOPID baselines that reproduce published results within 6.5 %, the tuned controller lowers the mean overshoot by 73 % and 53 %, the settling time by 46 % and 31 %, the sinusoidal MSE by 91 % and 77 % and the derivative-kick torque by 79 %. All eight independent runs satisfied every limit. The same optimiser with an ITAE-only objective tracks better but breaks 7 of the 18 torque limits, which shows that the effort terms and limits are essential for an applicable controller. Future work will compare FOPSO-GWO with other tuning algorithms under identical conditions, add gravity, friction and parameter uncertainty, and validate the controller on hardware.

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